MARLET: Multi-Agent Retrieval for Learning-State-Aware Educational Tutoring

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Abstract— Large language models can generate fluent tutoring responses, but they need grounding that respects both instructional evidence and the learner's current state. Classical retrieval-augmented generation (RAG) grounds generation through a fixed retrieve-then-read controller: the query is matched against a corpus, and the generator reads the returned passages. That design is often insufficient for tutoring, where the same surface question can require different evidence, scaffolding, or rubric constraints for different learners. We propose MARLET (Multi-Agent Retrieval for Learning-State-Aware Educational Tutoring), which places a supervisory router, planner, diagnoser, rubric module, retriever, tutor, and critic around a fixed downstream tutor. MARLET first builds a tutoring brief from the sample, inferred learning state, and answer criteria; retrieval is invoked only when the route indicates that external evidence is needed. Under matched backbones, corpus, reranker, and response budget on TutorEval and EduBench, MARLET improves fixed-tutor quality metrics while using fewer tokens; on EduBench it also reduces latency and backbone-call count. These results support the central claim: educational grounding should be controlled by pedagogical state and task regime, not by the surface query alone.

I. Introduction

Large language models (LLMs) are increasingly used as educational tutors because they can generate explanations, diagnose errors, and provide feedback [1], [2]. However, an LLM tutor cannot rely only on parametric knowledge. Knowledge-grounded dialogue has long shown the value of external evidence [3]; educational responses similarly need grounding in textbook chapters, course notes, rubrics, worked examples, or learner profiles to avoid fluent but misaligned guidance.

Classical retrieval-augmented generation (RAG) has therefore become the default grounding mechanism [4]. In classical RAG, the query is embedded, the top-k most similar passages are retrieved, and the LLM generates a response conditioned on those passages. Thus, relevance depends heavily on retrieval quality and query representation [5].

Educational tutoring is more complex than factual question answering. A student's question is often only a partial signal: the appropriate response may depend on level, goals, weak points, recent confusion, dialogue, and likely misconceptions. We call this the learner's personalized learning state. For example, an ordinary-differential-equation query may require high-school procedural scaffolding or college-level modeling context. Therefore, tutoring retrieval must find evidence that is pedagogically appropriate for the learner, not merely topical.

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Classical RAG can include explicit learner-state text when supplied, but appending learner state, dialogue, grading criteria, and task constraints forces one query embedding to compress topic, level, misconception, instructional goal, and response constraints. As constraints grow, the query can blur search intent and return evidence that satisfies some needs while missing others.

Therefore, we introduce MARLET (Multi-Agent Retrieval for Learning-State-Aware Educational Tutoring). Instead of letting the retriever act as the top-level controller, a supervisory router identifies the tutoring regime and decides whether retrieval is needed. A planner scopes search intent, a diagnoser summarizes visible learning state, and a rubric module states answer criteria before the fixed tutor writes.

The experiments validate the framework rather than present a model leaderboard. We compare with classical retrieve-then-read RAG under shared backbone, corpus, reranker, response cap, tutor, and critic. Our validation on TutorEval and EduBench shows that the same tutor scores higher when fed context assembled by MARLET; on EduBench it also reduces latency and backbone calls. Contributions. (i) We propose MARLET, a learning-state-aware multi-agent retrieval framework with a supervisory router, planner, diagnoser, rubric module, conditional retriever, tutor, and critic. (ii) We formalize the router and retrieval gate as transparent, training-free controls over learner state, rubric constraints, and external evidence. (iii) We validate against classical RAG on two educational benchmarks under shared backbone, corpus, reranker, and response budget.

II. Related Work

Classical RAG fixes retrieval as the first controller: the query is matched to passages, and generation conditions on the returned evidence [4]. Selective and corrective RAG methods improve this controller. Self-RAG learns when to retrieve, generate, and critique; CRAG evaluates retrieved documents and attempts correction when retrieval is unreliable; Adaptive-RAG and TARG vary retrieval behavior by query complexity or retrieval utility [6]–[9]. RAG diagnostics further emphasize separating retrieval, ranking, and generation errors [10]. These methods address whether passages retrieved for a query are necessary, sufficient, or reliable, but they do not infer learner stage before retrieval, separate rubric constraints from topic evidence, or decide whether external passages are the right source of grounding for a tutoring decision.

Agentic RAG and multi-agent LLM systems move beyond a single retrieve-then-read step. ReAct, AutoGen, CAMEL, and MetaGPT show that tool use and role decomposition can improve complex task execution [11]–[14]. MA-RAG uses multiple agents for query disambiguation, evidence extraction, and answer synthesis on ambiguous or multi-hop QA tasks [15]. These systems demonstrate coordination benefits, but their roles are designed for general task execution or ambiguous question answering. They do not define pedagogical regimes, infer visible learner state, or bind the final response to rubric-style answer criteria, so the educational reason for retrieving, skipping retrieval, or prioritizing rubric context remains implicit.

Recent educational work shows that tutoring quality requires more than answer accuracy. AgentTutor builds multi-turn personalized learning with learner assessment and memory [2]; adaptivity studies show that LLM tutors are sensitive to omitted student-context features [16]; and AI-tutor evaluation emphasizes pedagogy, mistake diagnosis, guidance quality, and learner support [1], [17]–[20]. These works define tutoring requirements, but they do not formulate retrieval as learner-state-aware control: when to retrieve, what evidence to retrieve, and how much context to allocate to evidence rather than learner-state and rubric information.

III. Methodology

MARLET is a tutoring framework that turns one educational request into a grounded tutor response. Its design follows three principles. First, routing is explicit: the system decides whether the request needs evidence, rubric feedback, adaptive tutoring, or planning before retrieval is attempted. Second, agents exchange structured state rather than free-form conversation. Third, retrieval is conditional and scoped by the tutoring brief, not by the surface query alone.

A. Input, Output, and Budget

For one sample, MARLET first standardizes all visible fields into

$$x = (q, o, c, h, r, v, m). \tag{1}$$

Here, q is the student question or task prompt; o is an optional answer-option list; c is optional inline context such as a chapter excerpt; h is the recent dialogue history; r is an optional rubric or answer-criteria list; v is an optional image list; and m stores metadata used for logging or formatting. Empty fields are allowed, so the same representation covers factual questions, grading requests, feedback requests, and planning requests.

The pipeline returns

$$\mathrm{out}(x) = (y, \mathrm{route}, D, \mathrm{cit}, \mathrm{log}), \tag{2}$$

where y is the final tutor answer, route records the selected pedagogical regime and module switches, D is the final evidence bundle, cit contains citation identifiers from retrieved chunks, and log stores trace and cost information. The budget is

$$B = (K, Q, L, N),$$

where K is the maximum number of planner retrieval queries, Q is the maximum number of evidence chunks/tool calls, L is the character budget for packed evidence, and N is the response-token cap. Experiments fix K=3, Q=4, L=6000, and N=900.

TABLE I

Framework components and responsibilities.

Component	Responsibility
Supervisor	Assigns pedagogical regime and activates retrieval, rubric, and critic switches.
Planner	Converts the tutoring request into an operating plan and up to three scoped retrieval queries.
Diagnoser	Infers visible learning state from the prompt and dialogue, including level and likely misconceptions.
Rubric module	Summarizes explicit rubric items or default answer-quality criteria.
Retriever	Runs only when requested by the route or fallback gate, then reranks and packs evidence.
Tutor/Critic	Generates the final response and checks evidence markers, rubric coverage, and pedagogy.

B. Supervisor Routing

The supervisor begins with a routing blob:

$$b(x) = \mathrm{concat}(q, c, r, h), \tag{3}$$

where $\mathrm{concat}(\cdot)$ means that the question, inline context, rubric text, and dialogue turns are joined into one normalized string. Images remain available to the tutor, but they are not converted into the textual routing score.

The router uses five cue families $j \in \{\mathrm{ev}, \mathrm{co}, \mathrm{ru}, \mathrm{pl}, \mathrm{tu}\}$ for evidence, coordination, rubric, planning, and tutoring. These families are defined because each one controls a different downstream decision: evidence cues open retrieval, coordination cues indicate that multiple specialist agents should assemble the brief, rubric cues activate criterion checking, planning cues route to lesson sequencing, and tutoring cues activate learner-state adaptation. Each family has a keyword set W_j. Evidence cues include source, document, chapter, and verification; rubric cues include score, criteria, and comment; planning cues include lesson, sequence, and exercise plan; tutoring cues include student, hint, misconception, explain, and confused. Let $M_j(x)$ be the number of cues from W_j that appear in $b(x)$. The raw cue score is simply

$$a_j(x) = \frac{M_j(x)}{|W_j|}, \tag{4}$$

where $|W_j|$ is the number of cues in family j. The supervisor uses this normalized cue score directly:

$$s_j(x) = a_j(x). \tag{5}$$

Here $s_j(x)$ is the final control score for family j. The thresholds are fixed router hyperparameters, not probabilities or learned weights. We choose them in cue-count units: one evidence or coordination cue contributes about 1/12=0.08, one rubric or planning cue about 1/6=0.17, and one tutoring cue about 1/7=0.14. Thus τ_{mid} is set to 0.35 so a route needs a repeated cue pattern rather than one accidental word; τ_{plan} is set to 0.42 because a false planning route changes the whole answer structure; and τ_{fb} is set to 0.45 because fallback retrieval costs an extra pass.

The pedagogical regime $R(x)$ is selected in a fixed order:

$$R(x) = \begin{cases} \text{LP}, & s_{\text{pl}} \geq s_{\text{ru}},\ s_{\text{pl}} \geq s_{\text{tu}},\ s_{\text{pl}} \geq \tau_{\text{plan}}, \\ \text{RF}, & s_{\text{ru}} \geq s_{\text{tu}},\ s_{\text{ru}} \geq \tau_{\text{mid}}, \\ \text{AT}, & s_{\text{tu}} \geq \tau_{\text{mid}}, \\ \text{EG}, & \text{otherwise}. \end{cases} \tag{6}$$

LP, RF, AT, and EG mean lesson planning, rubric feedback, adaptive tutoring, and evidence-grounded reasoning. Planning is checked first because lesson sequencing should not be reduced to short feedback, and its threshold is higher because planning changes the whole response structure. Rubric is checked before tutoring because score and criterion fields impose hard output constraints; tutoring is checked next when student-state cues reach the moderate-support threshold; otherwise the sample is handled as evidence-grounded reasoning.

The retrieval gate implements the same selective-retrieval intuition studied in adaptive RAG: evidence is requested when the sample appears to need external grounding rather than being forced for every input [6], [8], [9]. The gate is

$$G(x) = \begin{cases} 1, & s_{\text{ev}} \geq \tau_{\text{mid}} \text{ or } R(x) = \text{EG}, \\ 0, & \text{otherwise}. \end{cases} \tag{7}$$

Thus retrieval runs only when the evidence score reaches τ_{mid} or when the selected regime is evidence-grounded. This favors recall for initial grounding. A fallback gate handles missed evidence cases with the stricter τ_{fb}:

$$G_{\text{fb}}(x) = \begin{cases} 1, & D_0 \text{ is empty and } s_{\text{ev}} \geq \tau_{\text{fb}}, \\ 0, & \text{otherwise}. \end{cases} \tag{8}$$

where D_0 is the evidence bundle from the first retrieval attempt. If $G_{\text{fb}}(x) = 1$, the system performs one additional retrieval pass using the raw question.

C. Parallel Agent Brief Construction

MARLET uses role decomposition for planning, diagnosis, and rubric analysis, following the broader finding that tool-using and multi-agent LLM systems benefit from separating responsibilities [11], [12]. Agents do not send open-ended messages to each other. They read and write a shared context. The initial context is

$$z_0 = (x, R(x), G(x), B). \tag{9}$$

From z_0, the planner, diagnoser, and rubric module run in parallel when their switches are active:

$$(p, \ell, u) = \big(A_{\text{plan}}(z_0), A_{\text{diag}}(z_0), A_{\text{rub}}(z_0)\big). \tag{10}$$

Here p is the planner output, consisting of a short operating plan and up to $K = 3$ retrieval queries; ℓ is the visible learner-state summary; and u is the rubric summary or default answer-quality checklist. Parallel execution means these agents all read z_0 and return typed outputs independently; the tutor never has to infer the plan, learner state, and rubric from one overloaded query string.

The planner starts its query set with q, adds a compact key-term query from longer tokens in q, adds a context-aware query when c is present, and adds a rubric-aware query when r is present. Duplicate queries are removed and the list is capped by K. The diagnoser reflects tutoring work showing that LLM tutor quality depends on student-context features [2], [16]. It infers only visible state from q and h: level defaults to intermediate, beginner or advanced cues can override it, dialogue cues can add misconceptions, and style cues can request step-by-step, concise, or analogy-based explanation. This is not a persistent student profile. The rubric module preserves explicit criteria when available; otherwise it supplies default criteria of correctness, clarity, scaffolding, and actionable feedback. The merged tutoring brief is

$$z_1 = (z_0, p, \ell, u). \tag{11}$$

D. Conditional Retrieval and Evidence Packing

MARLET contains retrieval, but retrieval is not the top-level controller and does not require a traditional dense vector database. When $G(x) = 1$, each planner query q_i searches a local hybrid text index. For a query q_i and candidate chunk d, the base retrieval score is

$$\text{base}(q_i, d) = w_{\text{lex}}\text{lex} + w_{\text{char}}\text{char} + w_{\text{svd}}\text{svd} + \beta. \tag{12}$$

Here lex is word TF–IDF similarity, char is character 3–5 gram TF–IDF similarity, svd is truncated-SVD similarity, and β is a small title/source boost. The fixed weights $(w_{\text{lex}}, w_{\text{char}}, w_{\text{svd}}) = (0.58, 0.17, 0.25)$ emphasize lexical evidence while keeping typo robustness and semantic smoothing. Candidates from all planner queries are merged, and duplicate chunks keep their highest score.

The reranker then scores each candidate against the original question:

$$\begin{aligned} \text{rank}(q, d) =& \alpha_{\text{base}}\text{base} + \alpha_{\text{tok}}o_{\text{tok}} + \alpha_{\text{title}}o_{\text{title}} \\ &+ \alpha_{\text{phr}}m_{\text{phr}} + \alpha_{\text{num}}o_{\text{num}} + \alpha_{\text{len}}b_{\text{len}}. \end{aligned} \tag{13}$$

Here o_{tok} is token overlap with q, o_{title} is title-token overlap, m_{phr} is an exact phrase-match flag, o_{num} is numeric overlap, and b_{len} rewards moderate chunk length. The α weights are fixed to $(0.55, 0.20, 0.10, 0.06, 0.05, 0.04)$ in the order shown. The learner summary ℓ affects retrieval through the route and planner queries, not these weights.

Evidence packing follows the relevance-diversity trade-off used in MMR-style selection [21]: the context should be useful, but repeated chunks should not consume the limited evidence budget. For each candidate chunk d, MARLET computes

$$\text{pack}(d) = \lambda\,\text{rank}(q, d) - (1 - \lambda)\,\text{repeat}(d). \tag{14}$$

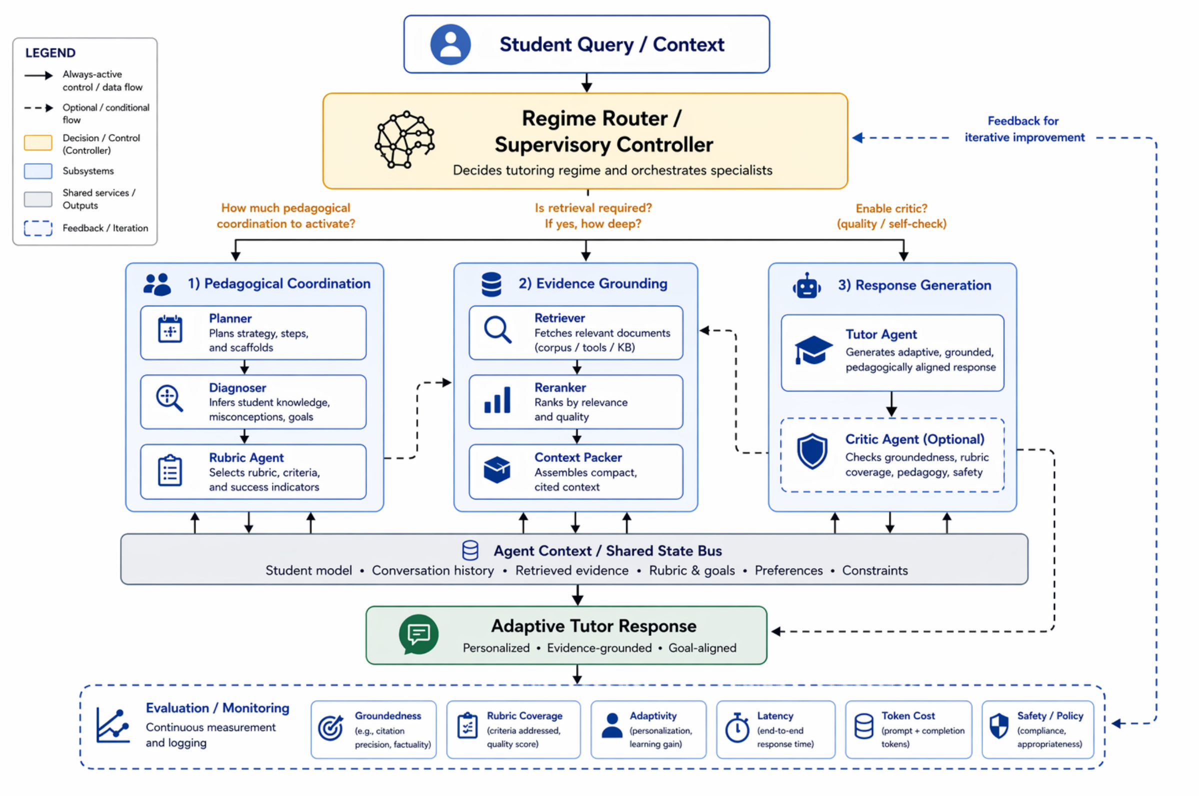

Fig. 1. Main framework of MARLET. The supervisory router scores the sample, assigns the pedagogical regime, and decides whether retrieval is needed before planner, diagnoser, and rubric modules assemble the tutoring brief.

Here $\mathrm{repeat}(d)$ is the largest sentence overlap between d and selected chunks, and λ=0.68 keeps relevance primary while penalizing duplicate evidence. The packer adds the remaining chunk with the highest $\mathrm{pack}(d)$ score until Q chunks or the L character limit is reached, forming evidence bundle D.

E. Tutor Generation and Critic Revision

The final tutor context is

$$z_2 = (z_1, D). \tag{15}$$

The tutor is the LLM-driven answer writer:

$$y_0 = A_{\mathrm{tutor}}(z_2). \tag{16}$$

Its prompt contains q, optional answer options o, recent dialogue h, learner state ℓ, plan p, rubric summary u, and either retrieved evidence D or inline context c. The tutor is instructed to be correct, pedagogically useful, concise but not terse, include a next step when appropriate, and cite retrieved evidence with document markers.

The critic performs one bounded revision pass:

$$y = \begin{cases} A_{\mathrm{crit}}(y_0, z_2), & I(y_0, z_2) > 0, \\ y_0, & I(y_0, z_2) = 0. \end{cases} \tag{17}$$

Here $I(y_0, z_2)$ is the number of issues found by the critic. Issues include missing citation markers when D is nonempty, uncovered rubric items from r, and answers that are too long for a beginner-facing state. If no issue is found, y_0 is returned unchanged; otherwise the critic rewrites once and the revised y becomes the final answer.

IV. Validation Experiments

We validate the framework on two public educational benchmarks. TutorEval [1] contains 834 expert-written science tutoring questions paired with long STEM chapters and teaching key points, so responses must explain, disambiguate, and stay chapter-grounded. EduBench [18] covers nine educational scenarios, including question answering, error correction, personalized support, emotional support, grading, teaching-material generation, and personalized content creation. Its rows contain prompts, scenario metadata, and reference model predictions; our adapter converts them into consensus-style supervision through score statistics and rubric terms.

The primary validation compares classical RAG with MARLET under matched Qwen3.5-4B infrastructure, reasoning budget 512, temperature 0.15, response cap 900, shared corpus, shared reranker, and fixed retrieval hyperparameters; a smaller Qwen3.5-27B-FP8 replication checks backbone sensitivity. Classical RAG runs retrieval first from the raw standardized prompt. If the prompt already contains a learner profile, student answer, dialogue, or rubric text, that visible information remains available to the baseline; it is not removed. What the baseline lacks is MARLET's explicit planner query construction, diagnosis, rubric brief, route decision, retrieval gate, and fallback logic. We report quality and cost together because MARLET changes context construction while keeping the infrastructure shared.

Evaluation is dataset-aware, and all reported quality scores are automatic proxies rather than official benchmark or human grades. Higher is better for quality metrics; lower is better for latency, tokens, and calls. Following RAG evaluation work [10], we separate answer overlap, rubric fulfillment, grounding, and resource cost. Token-F1 rewards overlap with the reference while balancing extra generated tokens against missed reference tokens. Rubric coverage checks whether each benchmark rubric item is actually expressed; near-verbatim coverage receives credit, and partial token overlap must include content-bearing words so that generic feedback is not over-rewarded. Latency is wall-clock time per sample; total tokens are prompt plus completion tokens; backbone calls count logged LLM requests.

For EduBench, EB12D averages 12 normalized checks in three groups: scenario adaptation, factual or reasoning behavior, and pedagogical application. These checks cover output-format compliance, supportive tone, relevance, use of scenario details, score agreement, domain and rubric grounding, reasoning quality, correction cues, clarity, motivation, personalization, and higher-order prompting. EduAlign is narrower: it only measures how close the extracted numeric score is to the benchmark's consensus score. Thus EduAlign can decrease even when explanatory feedback or rubric coverage improves. For TutorEval, tutor-hit measures expert teaching-point coverage. A key point receives full credit when it is explicitly stated or strongly covered, partial credit when the answer overlaps with enough of the key point, and no credit when the idea is absent. Off-profile metrics remain zeros in raw logs but are omitted from paper tables. The workload audit is a fixed weighted cost summary over tokens, retrieval queries, retrieved chunks, materialized agent outputs, and trace events; it is for efficiency inspection only and is not a training objective.

A. Framework Validation

Table II reports paired matched-sample 4B comparisons with bootstrap confidence intervals and permutation p-values. On TutorEval, MARLET raises Token-F1, tutor-hit, and rubric coverage while using fewer tokens and fewer backbone calls; latency trends lower but is not reliable (p=0.21). Table III reports the 27B replication.

B. Trade-offs

For TutorEval, the 27B paired CIs are $[0.0019, 0.0147]$ for Token-F1, $[0.0046, 0.0378]$ for tutor-hit, and $[0.0051, 0.0532]$ for rubric coverage; latency also trends lower with CI $[-1372, -297]$ ms. For EduBench, the 27B paired CIs are $[0.0616, 0.0778]$ for EB12D, $[0.1161, 0.1544]$ for rubric, and $[-0.0337, 0.0024]$ for EduAlign; cost CIs are $[-10658, -8985]$ ms, $[-2501, -2277]$ tokens, and $[-0.232, -0.146]$ calls. Classical RAG keeps a small EduAlign edge on EduBench, but it is not statistically reliable in this 27B slice.

TABLE II

4B quality and resource summary.

Data	Metric	RAG	Ours	Δ
TutorEval	Token-F1	0.112	0.115	$+0.004^\star$
	TE hit	0.938	0.957	$+0.019^{\star\star}$
	Rubric	0.919	0.944	$+0.026^{\star\star}$
	Latency (s)	51.90	49.11	-2.79
	Tokens	3975	3490	$-484^{\star\star\star}$
	Calls	1.84	1.56	-0.28
EduBench	EB12D	0.367	0.398	$+0.031^{\star\star\star}$
	Rubric	0.705	0.891	$+0.186^{\star\star\star}$
	EduAlign	0.166	0.126	$-0.040^{\star\star}$
	Latency (s)	19.83	13.90	$-5.92^{\star\star\star}$
	Tokens	4020	2226	$-1794^{\star\star\star}$
	Calls	1.97	1.48	$-0.49^{\star\star\star}$

$^{\star\star\star}p<10^{-4}$, $^{\star\star}p<0.01$, $^\star p<0.05$. Higher is better for quality; lower is better for latency, tokens, and calls. Runtime is mean wall-clock seconds/sample on the 4090 server.

TABLE III

27B quality and resource summary.

Data	Metric	RAG	Ours	Δ
TutorEval	Token-F1	0.264	0.272	$+0.008^\star$
	TE hit	0.897	0.918	$+0.021^\star$
	Rubric	0.861	0.891	$+0.030^\star$
	Latency (s)	41.47	40.65	$-0.82^{\star\star}$
	Tokens	6681	6730	$+49$
	Calls	1.99	1.99	-0.01
EduBench	EB12D	0.355	0.425	$+0.070^{\star\star\star}$
	Rubric	0.634	0.769	$+0.135^{\star\star\star}$
	EduAlign	0.541	0.526	-0.015
	Latency (s)	41.91	32.09	$-9.81^{\star\star\star}$
	Tokens	6105	3717	$-2388^{\star\star\star}$
	Calls	2.00	1.81	$-0.19^{\star\star\star}$

TutorEval uses n=400 paired samples; EduBench uses n=328 unique paired rows after duplicate public IDs in the 400-example slice are collapsed. $^{\star\star\star}p<10^{-4}$, $^{\star\star}p<0.01$, $^\star p<0.05$. Higher is better for quality; lower is better for latency, tokens, and calls. Runtime is mean wall-clock seconds/sample on the 4090 server.

MARLET can use more modules than classical RAG, so resource cost must be reported. Scoped planner queries and the retrieval gate reduce token use in 4B and still reduce latency in 27B, but the 27B TutorEval replication shows that better pedagogical coverage does not automatically mean fewer tokens. Thus MARLET is not a universal replacement for retrieval; it is a controller for deciding what grounding a tutoring sample needs.

V. Limitations

The study validates a framework rather than a deployable tutor: evaluation is automatic and English-only, and learner state is inferred from visible prompt/dialogue fields. Deployment requires privacy safeguards, bias checks, data governance, human oversight, and learning-gain studies.

VI. Conclusion

MARLET reframes educational grounding as supervised context construction rather than query-only passage lookup. With shared infrastructure, it improves fixed-tutor quality and efficiency over classical RAG by coordinating learner state, rubric constraints, and conditional evidence before generation. Future work should test stronger controller baselines, human and multi-turn outcomes, learned router calibration, profile-aware reranking, and privacy-preserving learner-state inference.

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